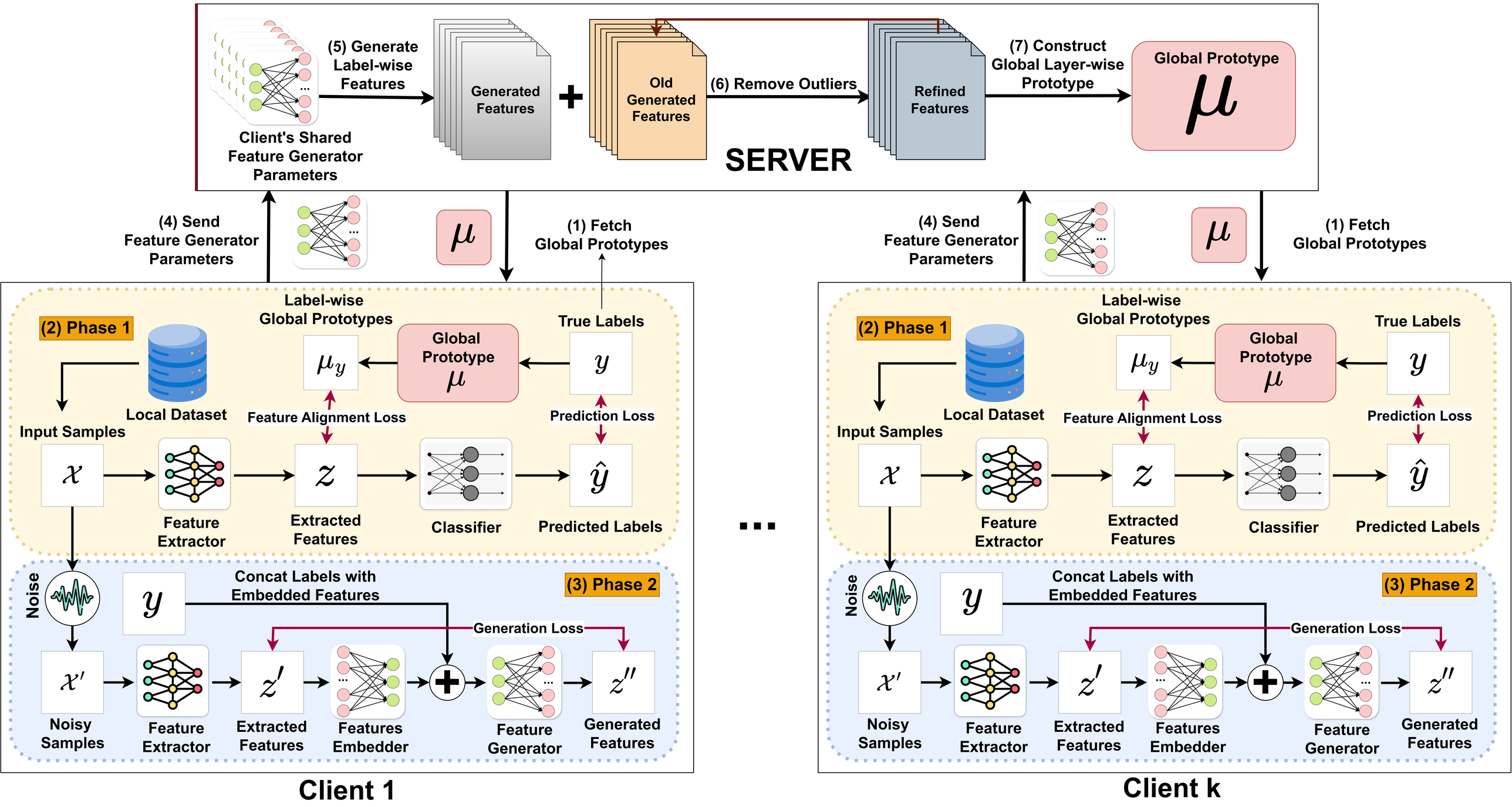

(5) Generate Label-wise Features
Client's Shared Feature Generator Parameters
Generated Features
Old Generated Features
(6) Remove Outliers
Refined Features
(7) Construct Global Layer-wise Prototype
Global Prototype
SERVER
(4) Send Feature Generator Parameters
(1) Fetch Global Prototypes
(2) Phase 1
Label-wise Global Prototypes
True Labels
Local Dataset
Input Samples
Feature Alignment Loss
Prediction Loss
Feature Extractor
Extracted Features
Classifier
Predicted Labels
Noise
Concat Labels with Embedded Features
(3) Phase 2
Generation Loss
Noisy Samples
Features Embedder
Feature Generator
Generated Features
Client 1
Client k